

CBSE Class 10 Mathematics — Some Applications of Trigonometry

Time Allowed: 3 Hours

Maximum Marks: 80

General Instructions:

1. This question paper contains *questions. All questions are compulsory.*
1. Questions are grouped into sections A through E.
2. Use of calculators is NOT permitted.

Section A: Multiple Choice & Assertion-Reason (1 mark each)

1. A tower stands vertically on the ground. From a point on the ground, which is 15 m away from the foot of the tower, the angle of elevation of the top of the tower is found to be 60° . The height of the tower is:
(A) (A) $5\sqrt{3}$ m
(B) (B) $10\sqrt{3}$ m
(C) (C) $15\sqrt{3}$ m
(D) (D) $20\sqrt{3}$ m
2. A kite is flying at a height of 60 m above the ground. The string attached to the kite is temporarily tied to a point on the ground. The inclination of the string with the ground is 30° . The length of the string is:
(A) (A) 60 m
(B) (B) 30 m
(C) (C) 90 m
(D) (D) 120 m
3. The angle of elevation of the top of a pole from a point on the ground 20 m away from the foot of the pole is 45° . The height of the pole is:
(A) (A) 10 m
(B) (B) 20 m
(C) (C) $20\sqrt{3}$ m
(D) (D) 40 m
4. If the length of the shadow of a tower is $\sqrt{3}$ times the height of the tower, then the angle of elevation of the sun is:
(A) (A) 30°
(B) (B) 45°

- (C) (C) 60°
 (D) (D) 90°
5. An observer 1.5 m tall is 28.5 m away from a chimney. The angle of elevation of the top of the chimney from her eyes is 45° . The height of the chimney is:
- (A) (A) 28.5 m
 (B) (B) 27 m
 (C) (C) 30 m
 (D) (D) 31.5 m
6. Assertion (A): The angle of elevation of the top of a tower from a point on the ground, which is 30 m away from the foot of the tower of height $30\sqrt{3}$ m, is 60° . Reason (R): In a right-angled triangle, $\tan(\theta) = \frac{\text{Opposite}}{\text{Adjacent}}$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 (C) (C) Assertion (A) is true but Reason (R) is false
 (D) (D) Assertion (A) is false but Reason (R) is true
7. Assertion (A): If the length of the shadow of a vertical pole is equal to its height, then the angle of elevation of the Sun is 45° . Reason (R): The angle of elevation is defined only when the object is taller than the observer.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 (C) (C) Assertion (A) is true but Reason (R) is false
 (D) (D) Assertion (A) is false but Reason (R) is true
8. Assertion (A): The angle of depression of an object from a point h meters above the ground is always equal to the angle of elevation of that point from the object on the ground. Reason (R): The angle of elevation and depression are alternate interior angles for parallel horizontal lines.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 (C) (C) Assertion (A) is true but Reason (R) is false
 (D) (D) Assertion (A) is false but Reason (R) is true

9. Assertion (A): If a ladder of length 10 m reaches a wall at a height of 5 m, the angle made by the ladder with the ground is 30° . Reason (R): $\sin(\theta) = \frac{\text{Opposite}}{\text{Hypotenuse}}$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true
10. Assertion (A): As an observer moves closer to a tower, the angle of elevation of its top increases. Reason (R): $\tan(\theta)$ increases as θ increases from 0° to 90° .
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true

Section C: Short Answer I (3 marks each)

1. A ladder 15 m long makes an angle of 60° with the wall. Find the height of the point where the ladder touches the wall.
2. The angle of elevation of the top of a tower from a point on the ground, which is 30 m away from the foot of the tower, is 30° . Find the height of the tower.
3. A kite is flying at a height of 60 m above the ground. The string attached to the kite is temporarily tied to a point on the ground. The inclination of the string with the ground is 60° . Find the length of the string.
4. A 1.5 m tall boy is standing at some distance from a 30 m tall building. The angle of elevation from his eyes to the top of the building increases from 30° to 60° as he walks towards the building. Find the distance he walked towards the building.
5. From a point on the ground, the angles of elevation of the bottom and top of a transmission tower fixed at the top of a 20 m high building are 45° and 60° respectively. Find the height of the tower.

Section D: Long Answer (4-5 marks each)

1. A tower stands vertically on the ground. From a point on the ground, which is 15 m away from the foot of the tower, the angle of elevation of the top of the tower is found to be 60° . Find the height of the tower.

2. A 1.5 m tall boy is standing at some distance from a 30 m tall building. The angle of elevation from his eyes to the top of the building increases from 30° to 60° as he walks towards the building. Find the distance he walked towards the building.
3. From the top of a 7 m high building, the angle of elevation of the top of a cable tower is 60° and the angle of depression of its foot is 45° . Determine the height of the tower.
4. A kite is flying at a height of 60 m above the ground. The string attached to the kite is temporarily tied to a point on the ground. The inclination of the string with the ground is 60° . Find the length of the string, assuming that there is no slack in the string.
5. The angle of elevation of the top of a chimney from the top of a tower is 30° and the angle of depression of the foot of the chimney from the top of the tower is 60° . If the tower is 40 m high, find the height of the chimney.
6. A straight highway leads to the foot of a tower. A man standing at the top of the tower observes a car at an angle of depression of 30° , which is approaching the foot of the tower with a uniform speed. Six seconds later, the angle of depression of the car is found to be 60° . Find the time taken by the car to reach the foot of the tower from this point.
7. The angle of elevation of the top of a tower from a point on the ground is 30° . On walking 40 metres towards the tower, the angle of elevation changes to 60° . Find the height of the tower and the distance of the initial point from the tower.
8. A vertical tower stands on a horizontal plane and is surmounted by a vertical flagstaff of height 6 metres. At a point on the plane, the angle of elevation of the bottom and top of the flagstaff are 30° and 60° respectively. Find the height of the tower.
9. From the top of a 7 m high building, the angle of elevation of the top of a cable tower is 60° and the angle of depression of its foot is 45° . Determine the height of the tower.